

Referee Report

Manuscript: “Skill-Replacing Technology and Bottom-Half Inequality”

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Summary of the contribution

The paper reframes routine-biased technological change (RBTC). In the canonical account (Autor–Levy–Murnane, 2003; Acemoglu–Autor, 2011), technology substitutes for routine *workers*, and its wage effect within a routine occupation is skill-neutral. Danieli instead lets technology substitute for the *skill used in routine tasks*, so that the return to skill *within* routine occupations falls. The mechanism is disciplined by a single cross-partial, $\varepsilon = \partial^2 \log \varphi_R / \partial \theta \partial \tau$, whose sign nests the three paradigms: skill-replacing ($\varepsilon < 0$), skill-neutral ($\varepsilon = 0$), and skill-enhancing ($\varepsilon > 0$) (Theorem 1). The object is micro-founded two ways—as task automation in which the type of RBTC is governed by the skill-intensity of the automated task (Appendix B.1), and as a CES aggregator in which it is governed by the skill–technology elasticity (Appendix B.2)—and extended to multidimensional skills (Appendix B.3). The model delivers a genuinely non-monotonic, two-phase prediction for bottom-half inequality and, in doing so, rationalizes three facts that skill-neutral RBTC cannot: (i) why the relative decline was concentrated at the *median* even though routine workers are dispersed across the whole bottom half; (ii) why that decline halted around 2000 while routine employment kept falling; and (iii) why *average* routine wages barely moved despite large employment losses.

Empirically the paper does two things. First, it derives and tests two new predictions—a falling return to skill and a falling *level* of skill in routine occupations—using an interactive fixed-effects model (IFEM) on the PSID (1980–2017), instrumenting the noisy worker fixed effect with years of schooling. It finds a sharp, sustained decline in the routine return to skill (Figure 4: $\log \alpha_R$ falls by more than 0.7, roughly a 50% reduction from its 1987 peak) and a steady fall in average routine skill that pushes it below the manual level by around 2015 (Figures 6–8). Second, it introduces a *skewness decomposition*—the third moment of log wages split into within, between, and covariance components—which sidesteps the ignorability assumption underlying standard decompositions. On CPS-ORG data it attributes 93% of the rise in skewness to occupational structure and $\sim 78\%$ to the covariance component (the correlation between an occupation’s wage level and its internal inequality)—the signature of SR-RBTC rather than of a skill-neutral price shock, which would load on the between component. The decomposition is a contribution of independent interest and is shipped as an R package.

Overall assessment

This is a strong, original paper. The core idea is simple, unifying, and—once stated—compelling: a single sign restriction reorganizes a large and previously untidy set of polarization facts, and the two-phase, non-monotonic prediction is a real advance over the monotone comparative statics of skill-neutral RBTC. On top of the theory, the paper offers *two* methodological contributions that will outlast the specific application: an IV-corrected IFEM for a setting with short panels, and the skewness decomposition. The writing is clear and the engagement with alternative explanations (institutions, minimum wage, unions, trade, service-demand shocks, rents, multi-skill models) is

unusually thorough. I expect the paper to be publishable after a focused revision.

The paper is transparent that it does not identify a *causal* effect of technology—there is no exogenous variation in τ —and that its claim is instead that a rich battery of facts is *jointly consistent* with SR-RBTC and hard to reconcile with the skill-neutral benchmark. I evaluate it on those terms, which I think are the right ones. My comments below are therefore aimed at strengthening the identification narrative and the robustness of the headline magnitudes rather than at the contribution itself; none of them, in my view, threatens the core result. They are ordered roughly by importance.

Main comments

1. Strict exogeneity of mobility is in tension with the model’s own mechanism. The IFEM’s consistency rests on strict exogeneity of occupational assignment (Eq. 28): the wage residual must be mean-independent of the full occupation history. But the central economic force in the model is precisely *selective* mobility—high-skill routine workers optimally exit as their comparative advantage shifts (Theorems 3 and 5). Selection on comparative advantage is a textbook violation of the assumption the estimator needs. The sensitivity analysis (Appendix D.2, following Andrews et al., 2017) is a valuable step and shows the *within*-routine trend is driven almost entirely by routine stayers, with a bounded bias from a trending schooling–error correlation. My concern is that this bounds one channel (the instrument–error covariance) but not Roy-style selection on the *transitory* wage component: if workers leave routine jobs when their idiosyncratic routine wage draw is poor, the wage–skill gradient estimated among stayers is a selected object even with θ_i held fixed. I would like the author to (a) state this tension explicitly, and (b) probe it—e.g. a selection/Heckman-style check on the stayer sample, a comparison of the $\alpha_{R,t}$ trend for cohorts with differing exit propensities, or bounds that allow the exit rule to depend on the transitory shock.

2. The incidental-parameters correction and inference in the IFEM. The worker effect θ_i is estimated from a handful of observations, so $\hat{\theta}_i$ is a noisy, inconsistent estimate and its use as a regressor induces attenuation—the incidental-parameters problem the paper correctly flags (Lancaster, 2000; Bound et al., 1994). This is exactly the difficulty that the leave-out / limited-mobility-bias literature on two-way fixed-effects models was built to handle, and the paper would be strengthened by connecting to it: Andrews, Gill, Schank & Upward (2008) on the direction and size of the bias, and Kline, Saggio & Sølvesten (2020) on leave-out estimation of variance components that are functions of estimated fixed effects. Three concrete requests. (a) *Relate* the schooling-IV correction to a leave-out estimator: does the instrument remove the entire bias in α_{jt} , or only the classical-measurement-error piece, leaving the correlated-error term (the second term in the D.1 probability limit) untouched? (b) *Inference*: the main text does not say how the standard errors on α_{jt} are computed given that $\hat{\theta}_i$ is a generated regressor entering nonlinearly. A bootstrap over workers, or an analytic correction, should be described and the resulting confidence bands shown on Figures 4–5; the sharpness of the trend is doing a lot of work and readers need its precision. (c) A short Monte Carlo—simulate the model, estimate the IFEM with and without the IV—demonstrating that the procedure recovers a known α under short panels would be persuasive and is cheap.

3. A single instrument for a single skill index. Identification of the return-to-skill trend hinges on schooling being uncorrelated with the wage residual conditional on θ_i (Eq. 10)—i.e. schooling affects wages only through the one-dimensional skill. This is strong in a paper whose

whole premise is that the returns to *different* skills move differently over time. If schooling also proxies, say, social or non-cognitive skills whose returns rose over this period (Deming, 2017), the estimated $\alpha_{R,t}$ trend is contaminated. The bound in D.2 is reassuring but leans on assumed magnitudes (a doubling of the residual-schooling coefficient over twenty years, $b \approx 0.001$) that read as somewhat ad hoc; I would motivate those numbers from data, and, if possible, show the trend is robust to a second instrument or to conditioning on an early cognitive-score proxy. The multi-skill robustness (Appendices D.3 and G) partly addresses this, but by construction the IV still recovers only the return to skills *correlated with schooling*; the paper should state that scope limitation where the main result is claimed, not only in the appendix.

4. From the covariance component to “return to skill”: tighten the bridge between the two exercises. The paper’s most novel empirical claim is that the covariance component—not the between component—drives polarization, and that this reflects a falling within-occupation return to skill. The counterfactual in Figure 12 (Eq. 16)—which holds occupation shares and means at their 1992 values and lets only the within-occupation variances move—reproduces essentially the entire rise in the covariance component; this convincingly establishes that the driver is within-occupation inequality *compression* in low-wage occupations, not shifts in occupation means or employment shares, and it is a clean result. What remains is the last inferential step—from “within-occupation inequality fell” to “the return to skill fell” (as opposed to the surviving workers simply becoming more homogeneous in skill). The IFEM is the tool that discriminates these, but it is estimated on a *different* dataset (PSID), at a *different* level of aggregation (three broad categories), than the decomposition (CPS-ORG, 3-digit). The headline mechanism therefore rests on bridging two exercises rather than on a single measurement. I would tie them together: for instance, show that the occupations where the IFEM α falls most are the same ones driving the counterfactual covariance in Figure 12, or estimate a coarser decomposition on PSID (and/or a coarser IFEM on CPS where feasible) so the two halves speak to the same units.

5. Skewness is tail-sensitive: report the robustness of the headline shares. The third moment is more sensitive to the tails than the variance, yet hourly wages are trimmed at 5% precisely because the tails are noisy—so the two design choices push against each other. The 93% and 78% shares are the paper’s quantitative punchlines and their dependence on (i) the trim and (ii) the 1992–2002 window should be shown, not asserted. A small figure plotting the three component shares as a function of the trim (say 1%–10%) and a table repeating the headline decomposition over alternative windows (the Autor–Dorn crosswalk period in Figure A12 is a start, but the coding-change caveat there is exactly why a clean robustness panel would help) would settle this. The R package presumably makes this inexpensive.

6. Calibrate the language of the abstract to what the decomposition measures. “SR-RBTC can explain 93 percent of the wage trends” (abstract) is stronger than what is shown: 93% of the rise in skewness is *occupation-related*, which is an upper bound on the share attributable to *all* occupation-based mechanisms—not only SR-RBTC but also, e.g., occupation-specific deunionization or the offshorability of tasks. The narrower $\sim 78\%$ covariance component is the SR-RBTC-consistent piece, and even that is consistent-with rather than uniquely-attributable-to. Appendix H’s occupation-vs-industry/education decomposition is the right kind of evidence and does a lot to narrow the field; I would simply have the abstract and introduction state the claim as “occupational structure accounts for 93%, of which the covariance component consistent with SR-RBTC is 78%,” which is both accurate and still striking.

Minor comments

- **Quantitative discipline on the turning point.** The theory is qualitative (signs of derivatives), and the mapping of the two phases to “late-1980s/1990s” and “2000s” is asserted, with the reversal point read off the same data it explains. A calibration in which the model *generates* the timing and magnitude of the ~ 2000 turnaround—even a stylized one—would convert a rationalization into a test. I flag this as a “would-strengthen,” not a requirement.
- **Sales classification.** The paper departs from prior practice by not coding sales as routine and notes sales has an abstract-like return to skill. Since occupational coding is central to every result, please show the main decomposition and IFEM figures under the conventional (sales-as-routine) definition side by side, rather than only asserting the results are unaffected.
- **Manual vs. routine wage/skill inversion.** By the end of the sample, routine workers have lower measured skill than manual workers yet still earn more; this is patched with an experience differential (Autor–Dorn, 2009). Because the θ –wage mapping is the backbone of the argument, this residual should be shown directly (e.g. the experience gap that reconciles the two) rather than relegated to a footnote.
- **Make the load-bearing fact a figure.** Puzzle (i) depends on “the highest-earning routine workers are located at the median of the overall distribution.” This is currently supported by citation; given its importance it deserves its own figure in the main text.
- **External validity.** The bottom-half fluctuation appears to be essentially a US (and Israel) phenomenon, attributed to weaker US labour-market institutions. This is a reasonable defense, but it also means the SR-RBTC footprint is hard to detect elsewhere; a sentence acknowledging the resulting limit on out-of-sample testing would be honest.
- **Cross-occupation rank evidence (Appendix G).** The rank correlations for category-switchers (0.35–0.53) are used to support one-dimensional skill, but movers are a selected sample and the correlations are not especially high; the θ_{ij} correlations (0.69–0.83) are more supportive. I would lead with the latter and note the selection caveat on the former.
- **Typos / presentation.** “was not applied to economic data” (Sec. 2.2); “orthong[on]al” (Appendix D.1); a few equation-rendering slips. Figure 1’s polarization series and the skewness series (Figure 10) could usefully appear closer together, since the paper’s opening argument is that they line up.

Recommendation

Minor revision. The contribution—the SR-RBTC reframing plus two portable methods—is novel and, in my judgment, publishable. The revision should focus on three things: (1) confront the tension between selective mobility and the IFEM’s strict-exogeneity assumption, and report generated-regressor inference for α_{jt} (comments 1–2); (2) tie the IFEM and the skewness decomposition to the same occupational units so the headline mechanism rests on one measurement rather than a bridge across datasets (comment 4); and (3) calibrate the language of the abstract, and add the trim/window robustness for the decomposition shares (comments 5–6). None of these requires new data or a redesign, and I would be glad to see a revised version.